Hawk and Hornet – Chemistry Lab

DO NOT WRITE ON THIS BOOKLET!!!!!

(or else your orbitals will be permanently hybridized)

Part 1: Multiple Choice = 30 points

1. The Earth's crust/mantle consists of numerous inorganic minerals with widely varying physical and chemical properties. One such mineral, perovskite (CaTiO3), is an important electronic component in solar cells and batteries. What is the formal scientific name of perovskite?

A. Titanium calcite
B. Titanium calcate
C. Calcium titanite
D. Calcium titanate

2. Building on question 1, what is the charge of Ti in perovskite?

A. +1
B. +2
C. +3
D. +4

3. Which of the following has the least number of molecules?
A. 16 g of H_2O
B. 21 g of C_3H_6
C. 84 g of CH_4
D. 47 g of N_2O

4. How many oxygen atoms are there in 15.0 grams of barium acetate?

A. 1.42×10^{23} oxygen atoms
B. 3.55×10^{23} oxygen atoms
C. 7.10×10^{23} oxygen atoms
D. 1.06×10^{24} oxygen atoms.

5. The following is what type of reaction:
$[Fe(C_5H_5)_2]BF_4 + NaB(C_6F_5)_4 \rightarrow [Fe(C_5H_5)_2]B(C_6F_5)_4 + NaBF_4$
A. Single displacement
B. Double displacement
C. Decomposition
D. Synthesis

6. Water decomposes into oxygen and hydrogen gas. How many liters of oxygen gas at STP can be made from decomposing 14.5 mL of liquid water (assume the density of water is 1.00 g/ml)?

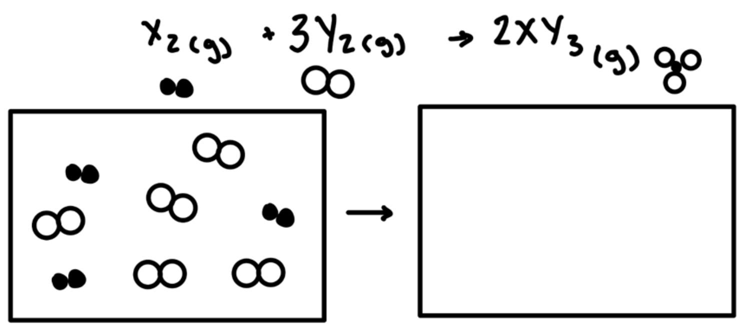
A. 9.02 L
B. 18.0 L
C. 22.4 L
D. 29.0 L

7. A student mixes 15.0 mL of 0.56 M sodium phosphate with 50.0 mL of 0.32 M barium nitrate. How many grams of precipitate will be formed?

A. 2.5 g
B. 3.2 g
C. 6.0 g
D. 7.2 g

8. Which of the following equations is correctly balanced?

A. $2\ Al_{(s)} + 3\ H_2SO_{4(aq)} \longrightarrow Al2(SO_4)_{3(s)} + 3\ H_{2(g)}$
B. $Sc_2O_{3(s)} + SO_{3(l)} \rightarrow Sc_2(SO_4)_{3(s)} + O_{2(g)}$
C. $Ca_3(PO_4)_{2(aq)} + 6\ H_3PO_{4(aq)} \rightarrow 4\ Ca(H_2PO_4)_{2(aq)}$
D. $TiCl_{4(s)} + H_2O_{(g)} \rightarrow TiO_{2(s)} + 2\ HCl_{(g)}$



9. The reaction pictured above occurs, which of the following is correct? (select all that apply)

A. 2.0 moles of XY_3 are produced.
B. 3.0 moles of XY_3 are produced.
C. 2.0 moles of X_2 remain unreacted
D. 1.5 moles of X_2 remain unreacted

10. What volume of 0.906 M hydrochloric acid solution can be prepared from the HCl produced by the reaction of 7.3 g of NaCl with excess sulfuric acid?

$$NaCl_{(s)} + H_2SO_{4(l)} \longrightarrow HCl_{(g)} + NaHSO_{4(s)}$$

A. 138 mL
B. 1380 mL
C. 69 mL
D. 250 mL

11. A 0.748 g sample of diprotic acid H_2A (MW 100.8 g/mol) is titrated with 28.2 mL of a NaOH solution. What is the concentration of the NaOH solution?

A. 0.526 M
B. 0.263 M
C. 1.05 M
D. 0.742 M

12. How many phosphate ions are in a sample of hydroxyapatite [Ca5(PO4)3OH] that contains 6.74 mg of oxygen?

A. 2.48×10^{19} phosphate ions
B. 5.85×10^{19} phosphate ions
C. 1.52×10^{19} phosphate ions
D. 3.25×10^{19} phosphate ions

13. Based on the mechanism below, what is the initial rate law for the reaction?

A. Rate = k'$[Cl_2][CH_4]$
B. Rate = k'$[Cl_2]^2[CH_4]$
C. Rate = k'$[CH_4][Cl]^2$
D. Rate = k'$[Cl_2]^{1/2}[CH_4]$

Step 1:
$$Cl_2(g) \rightleftharpoons 2Cl(g)$$
(fast equilibrium)

Step 2:
$$Cl(g) + CH_4(g) \longrightarrow CH_3(g) + HCl(g)$$
(slow)

14. Which of the following correctly represents k' given the reactions above?
a. $k' = (k_1/k_{-1})*k_2$
b. $k' = (k_1/k_{-1})^{1/2} *k_2$
c. $k' = (k_{-1}/k_1)*k_2$
d. $k' = (k_{-1}/k_1)^{1/2} *k_2$

15. A reaction is first order with respect to X and second order with respect to Y. What happens to the rate when [X] is doubled, and [Y] is halved?

A. The rate doubles
B. The rate is cut in half
C. The rate increases by a factor of 2
D. The rate remains the same

Suppose you're monitoring the decomposition of N_2O_5 and make the following observations:

$2\ N_2O_5\ (g) \rightarrow 4\ NO_2\ (g) + O_2\ (g)$

Time (min)	0	25	50	75	100
$[N_2O_5]$ (atm)	1.00	0.402	0.161	0.065	0.026

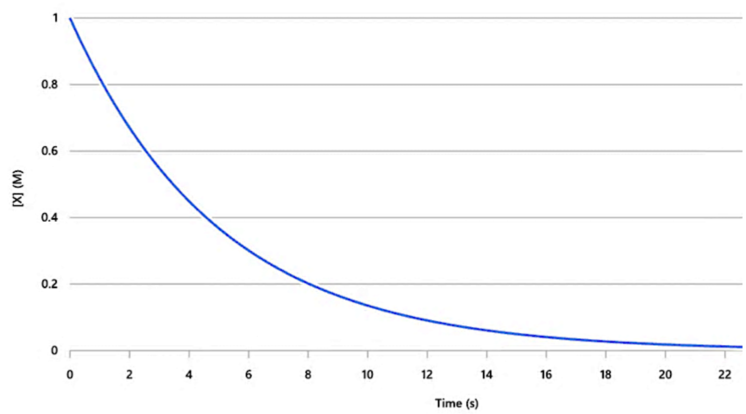
16. What is the average rate of disappearance of N2O5 between 25 and 50 minutes? (1pt)
A. 4.82×10^{-3} atm/min
B. 9.64×10^{-3} atm/min
C. 1.20×10^{-3} atm/min
D. 10.0×10^{-3} atm/min

17. At 75 minutes into the reaction, what percentage of products will have been generated?

A. 85.5 %
B. 78.1%
C. 95.4%
D. 93.5%

18. A catalyst is ______.
A. An agent that lowers the free energy of activation without being consumed
B. An agent that lowers the free energy of activation while being consumed in the process
C. An agent that lowers the free energy of the reaction product without being consumed
D. An agent that lowers the free energy of the reaction product while being consumed in the process

Use the graph below to answer questions 19-21.



19. If the concentration of X drops from 1.0 M to 0.25 M in 7 seconds, what is the approximate value of k?

A. 0.10 s^{-1}
B. 0.20 s^{-1}
C. 0.30 s^{-1}
D. 0.40 s^{-1}

20. Which of the following is the correct rate law for this reaction?

A. Rate=k[X]
B. Rate=k$[X]^2$
C. Rate=k
D. Rate=k$[X]^{-1}$

21. What is the half-life of chemical X?
A. 1.73 seconds
B. 3.47 seconds
C. 6.93 seconds
D. 12.50 seconds

$$Rate=k[A]^4[B]^2[C]^1$$

22. What is the correct unit of the rate constant, k, for this rate law?
A. L^6 / (mol^6 *sec)
B. L^7 / (mol^7 *sec)
C. L^5 / (mol^5 *sec)
D. mol^{-6} *sec^{-1} * L^{-6}

Use the reaction below to answer questions 23 and 24.

$$2Fe_{(s)} + 3CuSO_4{}_{(aq)} \rightarrow Fe_2(SO_4)_3{}_{(aq)} + 3\ Cu_{(s)}$$

23. A student places a 5.0-gram iron nail into a cup containing 100. mL of 0.35 M $CuSO_4$ solution. After the reaction has gone to completion, how many more grams of iron are left at the end of the reaction than grams of copper produced?

A. 0.9 g B. 1.2 g C. 2.0 g D. 3.0 g

24. If the student wanted to fully react the nail, how many mL of $CuSO_4$ would be needed to complete the reaction?

A. 270 mL B. 310 mL C. 380 mL D. 510 mL

Use the reaction below for questions 25-27.

The reaction below is the thermal decomposition of hydrogen peroxide. When this reaction is done mixed with soap, many people call it the "elephant toothpaste" demonstration because it looks like the toothpaste used to clean elephant tusks- probably not- that makes no sense- whatever moving on.

$$2H_2O_2{}_{(l)} \rightarrow 2H_2O_{(l)} + O_2{}_{(g)}$$

25. The density of 90% hydrogen peroxide can be estimated as 1.29 g/mL. How many liters of oxygen gas can be generated by decomposing 100.0 mL of 90% hydrogen peroxide at a pressure of 741.5 Torr and a temperature of 23.1^{o}C?

A. 21.3 L b. 31.8 L c. 47.3 L d. 56.4 L

26. The decomposition of H_2O_2 is first order with respect to H_2O_2. If the concentration of H_2O_2 is doubled, what happens to the rate?

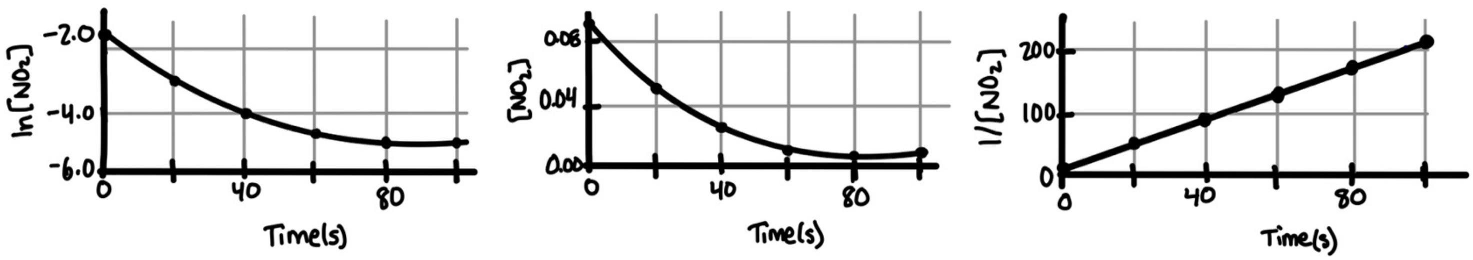
A. It doubles B. It triples c. It is halved d. It does not change

27. Why does the decomposition of H_2O_2 occur faster at higher temperatures?
A. Higher temperature decreases activation energy.
B. Higher temperature increases the number of molecules with sufficient energy to overcome activation energy.
C. Higher temperature changes the stoichiometry of the reaction.
D. Higher temperature decreases the frequency of collisions.

28. 10.0 mL of 0.10 M sodium sulfate is mixed with 15.0 mL of 0.85 M barium nitrate. Which option below lists the ions in the final solution from greatest to least molarity?

A. $NO_3^- > Na^{1+} > Ba^{2+} > SO_4^{-2}$
B. $NO_3^- > Ba^{2+} > Na^{1+} > SO_4^{-2}$
C. $NO_3^- > Na^+ > SO_4^{-2} > Ba^{2+}$
D. $Na^+ > SO_4^{-2} > Ba^{2+} > NO_3^-$

Use the graphs below for question 29 and 30.



29. The plots above are for the reaction: $NO_2 \rightarrow 2NO + O_2$. The rate constant for this reaction is most closely…

A. k=0.025 L mol^{-}1s^{-}
B. k=0.25 L mol^{-}1s^{-}
C. k=2.5 L mol^{-}1s^{-}
D. k=25 L mol^{-}1s^{-}

30. Does this reaction occur in 1 elementary step?
A. Yes- there is only one reactant.
B. Yes- the rate law is second order
C. No – the rate law does not match the reaction stoichiometry
D. Impossible to determine

Part 2: Written Response = 40 points

All calculations must be shown including starting equations, clear dimensional analysis with units, and your final answers must be boxed in with the correct units and significant digits.

1. Gold is a precious metal with a density of 19.32 g/cm^3 and is prized for its use in jewelry because it is chemically inert. Gold is often measured in karats with 18 karat gold being made of 75.0% pure gold. As of November 2025, the price of 18 K gold is $91.84 per gram.

a. **(7 pts)** If you have a tennis ball with a diameter of 6.54 cm made of solid 18 K gold, how many gold atoms would exist in the tennis ball?

b. **(3 pts)** How much would the tennis ball cost if it were to be made entirely of solid 18 K gold?

During World War II, chemist George de Hevesy dissolved the Nobel Prize medals of German physicists Max von Laue and James Franck in aqua regia to hide them from the Nazis. After the war, he recovered the gold from the solution and the medals were recast by the Nobel Foundation. The reaction for aqua regia is shown below.

$$\mathbf{Au_{(s)} + 3HCl_{(aq)} + HNO_{3\,(aq)} \rightarrow AuCl_4^-{}_{(aq)} + NO_{(g)} + 2H_2O_{(l)}}$$

c. **(7 pts)** Aqua regia is typically made from a solution which mixes concentrated 12 M 1HCl and 16 M HNO_3. What would the total volume of aqua regia solution that would be needed to completely react the solid 18 K gold tennis ball?

2. A student wants to determine the mass percent of potassium carbonate (K_2CO_3) in an antacid tablet, which contains potassium carbonate as well as several other soluble fillers. They do so by dissolving the tablet in water, and then reacting the contents with an excess of 0.50 M strontium chloride solution according to the reaction below.

$$\mathbf{SrCl_{2\,(aq)} + K_2CO_{3\,(aq)} \rightarrow SrCO_{3\,(s)} + 2KCl_{(aq)}}$$

Data is taken and collected in the table below.

Mass of tablet (g)	3.90 g
Mass of filter paper (g)	1.22 g
Mass of filter paper + precipitate (g)	2.81 g
Volume of $SrCl_2$ (mL)	35.00 mL

(8 pts) Calculate the mass percent of potassium carbonate in the original tablet.

3. The net ionic equation below shows the process by which iodine can be generated in the laboratory through the reaction of hydrogen peroxide in an acidified solution of potassium iodide.

$$H_2O_2(aq) + 2I^-(aq) + 2H^+(aq) \rightarrow I_2(aq) + 2H_2O(l)$$

A 20.00 mL sample of 0.40 M KI reacts completely with excess hydrogen peroxide. The iodine produced is collected and measured.

a. **(3 pts)** Calculate the number of moles of $I^-_{(aq)}$ present in the reaction vessel before the reaction occurs.

b. **(3 pts)** Calculate the number of moles of $I_{2\,(aq)}$ produced after the reaction takes place.

T	$[I^-]$	$[H_2O_2]$	$[H^+]$	Rate (M/s)
1	0.0100	0.00500	0.050	1.20×10^{-6}
2	0.0300	0.00500	0.050	3.60×10^{-6}
3	0.0300	0.0100	0.050	7.20×10^{-6}

c. **(4 pts)** Using the information in the table, determine the order of the reaction with respect to each of the following. Justify your answers with math or words.

i. I^-
ii. H_2O_2

d. In a separate experiment, it is determined that the reaction is first order with respect to H^+ in addition to your results from di and dii. Using this information and your answers to part e above, complete the following:

i. **(2 pts)** Write the rate law for the reaction.

ii. **(3 pts)** Calculate the value of the rate constant, *k*, for the reaction including the appropriate units.

Student Names: ______________________________ **Team Name/Number:** ____________________

All calculations must be shown including starting equations, clear dimensional analysis with units, and your final answers must be boxed in with the correct units and significant digits.

1a. (7 pts)
1b. (3 pts)
1c (7 pts)

2a. (8 pts)

3a (3 pts)

3b (3 pts)

3ci (2 pts)

3cii (2 pts)

3di. (2pts)

3dii (3 pts)

Hawk & Hornet Chem Lab ________/ 15 points

Student Names: ____ **Team Name/Number:** ________________________

Overview: In this lab we will react green food coloring with sodium hypochlorite (bleach – NaClO). We will make the bleach an excess reactant so that its concentration remains relatively constant throughout the experiment so we can determine the order of the reaction with respect to the green food coloring.

Reaction: $C_{37}H_{34}N_2O_{10}S_3Na_2$ + NaClO → Products that are colorless
^ Green food coloring

Procedure:

1. Confirm that the spectrophotometer is set for a wavelength of 635 nm. Zero the instrument using the correct procedure as outlined on the direction sheet.

2. Measure 50. mL of food coloring and water solution into your beaker.

3. Measure 3 mL of bleach using the 10 mL graduated cylinder.

READ ALL OF THE NEXT DIRECTIONS BEFORE PROCEEDING.

4. Make sure you are organized to collect the absorbance every 10 seconds.

5. When ready, add the bleach to the water/food coloring mixture. Stir it so the bleach gets distributed.

6. ***Quickly*** add some of the solution to the cuvette and place the cuvette into the spectrophotometer with the cloudy side facing you and the clear side facing left to right **without the lid**. Begin to collect data immediately.

Data: (3 pts)

Time (sec)	Abs	ln(abs)	1/Abs	Time (sec)	Abs	ln(abs)	1/Abs
5.00				90.00			
10.00				100.00			
20.00				110.00			
30.00				120.00			
40.00				130.00			
50.00				140.00			
60.00				150.00			
70.00				160.00			
80.00				170.00			

The following questions should be answered below.

1. (2 pts) Identify the order of reaction and write the rate law that best fits the data.

2. (4 pts) Determine the value of k, and the units for k.

3. (2 pts) Determine the half-life for this reaction.

4. (2 pts) Given a starting food coloring concentration of 0.50 M, determine the food coloring concentration after 1 minute 11 seconds have passed.

5. (2 pts) ________ A student repeats the experiment but uses a wavelength greater than the wavelength of maximum absorbance does this cause the data to be…

   a. Too high   b. Too Low   c. Not impacted

Multiple Choice KEY

1. D
2. D
3. B
4. A
5. B
6. A
7. A
8. A
9. A & C
10. A
11. A
12. B
13. D
14. B
15. B
16. B
17. D
18. A
19. B
20. A
21. B
22. A
23. A
24. C
25. C
26. A
27. B
28. A
29. C
30. C